Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

___show ip route ospf_______________________________________ 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/0_________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __65_______ 1290 0

Does R2 show up as an OSPF neighbor on R1? __yes______ 2

Does R2 show up as an OSPF neighbor on R3? ____no____ 2

What does this information tell you?

This shows that R2 forms an OSPF neighbor relationship with R1 but not with R3.

Therefore, routing updates between R3 and R2 must go through R1.

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/1______ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

____64 (serial link) + 1 (GigabitEthernet) = 65______________________________ 2

Is R2 listed as an OSPF neighbor to R3? ___yes____ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because modern networks use links faster than 100 Mbps.

Changing the reference bandwidth allows OSPF to calculate more accurate costs for high-speed links such as Gigabit or 10-Gigabit Ethernet.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The OSPF route cost is calculated by adding the cost of each outgoing interface along the path.

For example, the route to 192.168.3.0/24 includes the cost of the serial interface and the Gigabit Ethernet interface. The sum of these interface costs becomes the accumulated route cost

2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

845 Because both serial interfaces now have the same bandwidth setting (128 kbps), so their OSPF cost becomes equal. Therefore, R1 calculates two equal-cost paths with a total cost of 845.

0

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

4

After applying "ip ospf cost 1565" to R1's S0/0/1, the direct path from R1 to 192.168.3.0/24 becomes: • Via S0/0/1 directly to R3: cost 1565 (S0/0/1) + 1 (R3 G0/0) = 1566. The alternate path via R2 is: • Via S0/0/0 to R2: cost 781 (S0/0/0) + 781 (R2 S0/0/1) + 1 (R3 G0/0) = 1563. Since 1563 < 1566, OSPF selects the path through R2 as the best (lowest-cost) route. The ip ospf cost command directly overrides the bandwidth-calculated cost and forces OSPF to prefer the multi-hop path via R2 over the direct serial link to R3.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   The router ID uniquely identifies each router in the OSPF domain.

   Controlling it helps ensure stable neighbor relationships and easier network management. 4

2. Why is the DR/BDR election process not a concern in this lab?

   Because most connections in the topology are point-to-point serial links, and DR/BDR election only occurs on broadcast networks. 4

3. Why would you want to set an OSPF interface to passive?

   To prevent sending OSPF hello packets on LAN interfaces while still advertising the network. 4

   This reduces unnecessary routing traffic and improves security.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of areas (especially Area 0 backbone) allows OSPF to divide large networks into smaller areas, making the network more scalable. 4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   ______network 10.0.0.0 0.255.255.255 area 0__________________________ 4

   ____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   OSPF routing will be used. Routing decisions are based on administrative distance (AD) rather than metrics of the protocol itself. AD is used to compare the routes of different routing protocols: •OSPF AD = 110•EIGRP AD = 90•RIP AD = 120 EIGRP has the lowest AD(90), so EIGRP routes will be installed in the routing table and used for forwarding. When EIGRP routes are present, OSPF and RIP routes serve as backups but are not in the routing table.

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